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P. A. M. Dirac - Session I

April 1, 1962

Interviewed by: Thomas S. Kuhn and Eugene Wigner

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Abstract:

Part of the Archives for the History of Quantum Physics oral history collection, which includes tapes and transcripts of oral history interviews conducted with circa 100 atomic and quantum physicists. Subjects discuss their family backgrounds, how they became interested in physics, their educations, people who influenced them, their careers including social influences on the conditions of research, and the state of atomic, nuclear, and quantum physics during the period in which they worked. Discussions of scientific matters relate to work that was done between approximately 1900 and 1930, with an emphasis on the discovery and interpretations of quantum mechanics in the 1920s. Also prominently mentioned are: Niels Henrik David Bohr, Max Born, Boyland, Louis de Broglie, Johannes Martinus Burgers, Paul Ehrenfest, Ralph Fowler, Peter Fraser, Werner Heisenberg, Ernst Pascual Jordan, Cornelius Lanczos, Edward Arthur Milne, Wolfgang Pauli, David Robertson, Ernest Rutherford, Erwin Schrödinger, John Joseph Thomson, Hermann Weyl; University of Cambridge, Delta Squared V Club, Kapitsa Club, Kbenhavns~Universitet, Merchant Venturer's School in Bristol, University of Bristol, and Universität Göttingen.

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Dirac:

I started my education at Bristol. It was my home town. My father was a master in one of the schools there. He taught French. I went to the same school. When I completed that I moved on to the faculty of engineering of the University of Bristol. It was in the same building. So I continued my work in the same building. I got pushed on pretty fast because the First World War was on at the time. That meant that the young men were called up and there were plenty of vacancies. I started the University when I was 16. I got a degree in electrical engineering at 19. I looked around for an engineering job, but we were just at the depression at that time and I couldn't get any. My professor of engineering said that instead of just hanging around doing nothing, I should do some research in engineering. I had just started that for a few weeks, when he suggested that I take a course in mathematics at the University of Bristol. That involved going to a different building.

Kuhn:

How much mathematics had been in the engineering curriculum?

Dirac:

Differential equations — as far as that. I mean they were not concerned With mathematical rigor; they were concerned with getting results.

Kuhn:

Were transient circuit techniques — this sort of thing — involved?

Dirac:

Yes. I think that this engineering education has influenced me very much in making me learn to tolerate approximations. My natural feelings were to think that only an exact theory would be worth considering. Now, engineers always have to make approximations. I learned that even a theory based on approximations could be a beautiful theory. I rather got to the idea that everything in nature was only approximate, and that one had to be satisfied with approximations, and that science would develop through getting continually more and more accurate approximations, but would never attain complete exactness. I got that point of view through my engineering training, which I think has influenced me very much. As a result of that I haven't been much interested in questions of mathematical logic or any attempts to form an absolute measure of accuracy, an absolute standard of reasoning. I feel that these things are just

not important, that the study of nature through getting ever, improving approximations is the profitable line of procedure.

Kuhn:

Had you been heading for engineering right along?

Dirac:

I think so. I had an elder brother who also took engineering, and I tended to copy him. I didn't have much initiative on my own. That path was rather well set out for me, and I didn't know very well what I wanted.... The man I was in closest contact with in my engineering work was Professor David Robertson, who was a professor of electrical engineering. I was very much influenced by him. I was very much impressed by the need to have foresight to avoid accidents. He also arranged things to show the mathematical beauty, I might say, of many of the calculations that one had to make. He was a man who was paralyzed through polio, I think. He moved around in a wheelchair everywhere. He had to organize his whole life to be able to manage with that handicap. He seemed to do it pretty well. I then moved over to the mathematic department at Bristol University and came under the influence of Professor Hasse, the professor of applied mathematics was, and Mr. Fraser, a lecturer in pure mathematics. Dr. Fraser was an extremely good lecturer. He introduced me to the ideas of mathematical rigor, and he was able to make the subject interesting. I spent two years on his course in mathematics at Bristol University, then I went to Cambridge as a research student. By going to Cambridge as a research student, I didn't need to learn Latin. Latin was a compulsory subject at Cambridge, but they never taught it at my school. I should say that the school I went to as a secondary school was an extremely good scientific school. Most of my time there was spent learning science and mathematics and modern languages..... At an early stage in secondary school I was allowed to do mathematical reading by myself at a more advanced level than the rest of the class. I was just told which books to read and I studied them by myself.

Kuhn:

What sort of books did you read?

Dirac:

Books on differential and integral calculus. I think Edwards was the author, but I'm not very sure. I don't think it matters very much which books one reads, if one has reasonably good books.

Kuhn:

Did you get on to other parts of mathematics at that point, besides calculus?

Dirac:

Just geometry and algebra.... I think all the time I picked up my mathematics more by working by myself than from lectures. I don't seem to be able to pick things up very much from a lecture because I like to jump forward and jump back again and jump forward and back, continually. One can't very well do that if one is listening to an ordered presentation like a lecture. When I go to lectures I usually just get stimulated to think on certain lines, and then maybe I think along those lines myself instead of listening to every word the lecturer says. I perhaps miss a good deal of the lecture for that reason and have to make it up later in my own reading or something. But all my learning in mathematics has been rather along those lines. And it still is like that.

Kuhn:

To what extent did books outside of the curriculum also have this same role of starting problems, starting ideas?

Dirac:

I didn't have very many books available to me. I just had a few which I studied thoroughly.... I know I was always very interested in the fundamental problems of nature. I would spend much time just thinking about them. I remember I thought out for myself that there might be a connection between space and time coordinates such that when one changes one's time axis, one would also rotate one's space axis. But I knew nothing about the hyperbolic geometry at that time. I could see that it wouldn't work just with Euclidian space because going through big angles, you soon get into contradictions.

Kuhn:

How far did the science itself go in school? Were you again there encouraged to go further than the courses?

Dirac:

To some extent, but not as much as I would have liked to. I had the advantage that I was pushed up into a higher class than would normally correspond to my age just on account of the war. All the older students were sent off to war work and left the higher classes empty. That meant that all through my secondary schooling I got put into a higher class than would correspond to my age. It was an advantage from the point of view of learning things early.

Kuhn:

When ‘you did the two years of math at Bristol, what subjects did you then have to deal with?

Dirac:

I learned about the rigorous setting up of calculus. Previously, in the engineering faculty, one just learned it from the practical point of view, and one didn’t bother about getting things rigorous. I also learned differential geometry. That was Fraser. I found that a most interesting subject. Did I say differential geometry? I made a mistake, I should have said projective geometry. No, I did not do differential geometry at that stage. Projective geometry I found a most fascinating subject. One could get quite powerful results — theorems about straight lines and conies intersecting each other — just from elementary arguments about 1 to 1 correspondence. That appealed to me very much. All my work since then has been very much of a geometrical nature, rather than of an algebraic nature.

Kuhn:

Did you do abstract algebra?

Dirac:

No, I don’t know that that existed then. Did it? I did not do any of that. I learned a little of quaternion just by reading up on it myself. I got hold of Thomson’s textbook. Rather heavy- going. There was quite a lot of unnecessary heaviness in the beginning of that book, but I learned something about it.

Kuhn:

Were quaternion still a live method in British education, or were there people using them still at Cambridge or at Bristol?

Dirac:

At Bristol, no. At least so far as I know. I can’t remember how I first got hold of that interest myself. And not very much in Cambridge. But it is a subject that I find fascinating myself. I don’t think it is very much appreciated. I don’t think it is in general appreciated as much as it should be. * * *

Kuhn:

Did. you pursue the quaternion at all?

Dirac:

No, I didn't. When the war ended, there was tremendous interest in relativity. Previously we just hadn't heard about it at all, and then there was this tremendous interest.

Professor Broad, who is a professor of philosophy, gave a course of lectures on the subject, which I attended.... Broad was at Bristol. He talked about it largely from the point of view of a philosophy. I tried to appreciate it, but I did not get very much success in trying to appreciate philosophy. But I did learn from Broad the three plus signs and one one minus sign, which was the basis of the Einstein theory. I saw that it was really something new which I had never thought of in my speculations about relations between space and time

Kuhn:

But it tied to those speculations.

Dirac:

Well, it provided a way out of the difficulty. The difficulty that one got into contradictions when one made too big rotations....

Kuhn:

There had been little or no talk of relativity in your scientific education up to that point?

Dirac:

Not even the special theory. I hadn't heard about it.

Wigner:

How much did you see personally of teachers in those places? at is something I always wanted to ask you.

Dirac:

I went to lectures. Do you mean, about social contacts?

Wigner:

Not only social.

Dirac:

I think once or twice they invited me to their homes. I know David Robertson invited me to his home a few times. Fraser also has invited me to his home. Fraser wasn't married at that time. He married only later.

Wigner:

But did you see them personally for scientific discussions?

Dirac:

No. They did not have anything in Bristol corresponding to the supervision. Nor did the undergraduates in Cambridge. In the mathematics course the class was very small, it was only two people, a girl and, myself. If I remember — I can't at the moment. It may come back.

Wigner:

Did you talk to her a great deal about mathematics, about the subject?

Dirac:

No, I think we only met at the lectures and, separated afterwards, so far as I know.

Wigner:

You didn't invite her to tea?

Dirac:

Oh no. In fact I had no social life at all as a child.

Wigner:

You had no social life. Where did you stay, Paul? 1There did you eat, where did you sleep?

Dirac:

Well I lived with my parents, but nobody ever, came to our house, except a few pupils of my father who came from lectures. No one ever came for social purposes.

Wigner:

How much did you talk to your parents?

Dirac:

Very little. My father made the rule that I should only talk to him in French. He thought it would be good for me to learn French in that way. Since I found that I couldn't express myself in French, it was better for me to stay silent than to talk in English. So I became very silent at that time- - that started very early. My father and mother usually ate separately. My mother in the kitchen and father in the dining room I used to eat in the dining room with my father; and my brother and sister would eat in the kitchen with mother. That was a general rule. I think I would have preferred to be with the others, but there weren't enough chairs in the kitchen, and so I would eat in the dining room with my father.

Kuhn:

Were they younger or older than you?

Dirac:

My brother was older, and sister was younger.

Kuhn:

Was he very much older?

Dirac:

He was two years older. He committed suicide at the age of 24, which was a great shock to my family, of course.

Wigner:

You've never told me what the cause of it was, if you knew.

Dirac:

I suppose he was just very depressed. That kind of life, brought up without any social contacts must have been very depressing to him as it was to me. And having a younger brother who was brighter than he was must have depressed him also quite a lot.

Wigner:

Was it evident that you were brighter than he was?

Dirac:

Yes it was. For instance, I got a first class in my engineering degree and he got a third class.

Wigner:

Did he have a Job?

Dirac:

Yes, he got an engineering job in the Midlands, Coventry, for some time, and Wolverhampton. I think he seemed to like to come home, even though there wasn't much social life at home. The small amount of vacation that he would get he would immediately spend the time by cycling to Bristol and cycling back again in the end, spending as much time at home as he could.

Wigner:

Was he interested in any girls?

Dirac:

He did have a girlfriend, yes, but I don't know how close his relationships with her were.

Wigner:

What did she say when he died?

Dirac:

I don't know. My father did suggest that this girl should come and visit us, and my mother said, "Oh no, she mustn't, she might go after Paul." I was 22 years at that time, but my mother still thought I needed to be protected from girls. I rather resented it, but the result was that I never met her. There was some mystery connected with my brother's death. He left his job three months before he died, and what he did during those last three months, no one was able to find out. He didn't tell his landlady that he'd left his job. He continued to leave regularly in the morning and come home in the evening. His landlady did not know that he had left his job. He continued to pay his rent regularly. He just withdrew his savings, and when they were gone he killed himself. The

police made extensive inquiries, but they were unable to find out what he did during those last three months.

Wigner:

Did he leave his job of his own ... ?

Dirac:

Yes, he left it voluntarily.

Wigner:

Did you know that he left his job?

Dirac:

Oh no. We didn't correspond with each other. In fact we didn't talk to each other for a number of years.

Wigner:

Why not, Paul?

Dirac:

Well, for one thing we had to talk in French or we'd get into a row. That was one reason for not talking.

Wigner:

This French rule applied to your brother also?

Dirac:

Yes, and to my sister. He thought he would make us learn French in that way. My father was Swiss by birth... My mother was English. I think father also had an unhappy childhood. He ran away from home without telling his parents and went to England.. He got married without telling them. His parents never knew that he was married until some years later when he visited them with a couple of children. That's what I heard from my mother.

Kuhn:

Did they encourage your turning to science from engineering?

Dirac:

My father, yes, always encouraged me toward mathematics. He appreciated very much the importance of a good education, and he always encouraged me in that way. He did not appreciate the need for social contacts. The result was that in those days I didn't speak to anybody unless I was spoken to. I was very much an introvert, and I spent my time thinking about problems in nature. And, of course, when I first learned about relativity, that was a period of great excitement.

Kuhn:

You went right on with it from there?

Dirac:

Well, I got Eddington's book and studied that for some time

Wigner:

How does it come that Eddington had a book ready so soon if relativity was not known at all in England?

Dirac:

Well it wasn't known at Bristol. Eddington did get off very quickly. Eddington introduced the subject to England.

Kuhn:

These lectures of Broad's would have been immediately after the war?

Dirac:

I should think: about 1918. I'm not too sure of the exact date.

Kuhn:

This would have been before the eclipse expedition?

Dirac:

Yes, and of course the eclipse expedition did enhance that interest.

Kuhn:

Were there other things about relativity that you recollect as being particularly exciting?

Dirac:

Well the idea of curved space was introduced at that time also. We learned about the special theory and the general theory of relativity simultaneously.

Wigner:

Did the constancy of the velocity of light play any role in your way of thinking?

Dirac:

I didn't know that it was constant.

Kuhn:

You got the relativity in a fairly highly mathematical formulation at the time you first heard of it.

Dirac:

No, Broad didn't do very much mathematics. He didn't prepare me.... It needed some time to get used to Eddington, to get used to the tensor calculus.

Kuhn:

Were there people working hard at it, discussing it at all, at Bristol then? I mean was this a general excitement, or was this something unique?

Dirac:

Other students were. Just one of those early students with me I've kept in touch with — a man called Wilshire. I believe we used to talk about it together. He is living not far from Cambridge and visited me a few years back. You could get more information from him about it.

Kuhn:

What is his first name?

Dirac:

I think it is H. C., but I am not sure about that. He works at an aeronautical college near Bedford.

Kuhn:

What was your sense of change when you went from Bristol to Cambridge? Did you find the environment very different, the subjects more advanced?

Dirac:

Yes. First of all going away from home. I had been away from home previously during a long vacation. During my engineering course I went to (Rugby) for about a month or so to get some practical experience in engineering work at the B. T. H. works - - the British Thompson Houston Works in Rugby. My brother was also working there at the time, but we never met. If we passed each other in the street, we didn't exchange a word. I got some practical experience there working in an engineering factory. I did not please my employers very much, and they sent an unfavorable report back to my professor, Professor David Robertson.

Wigner:

How did you find out about this?

Dirac:

He told me. David Robertson, yes, he showed me the report.

Wigner:

What did the report say-?

Dirac:

It said I lacked keenness, and was slovenly.

Wigner:

You lacked keenness? And you are slovenly?

Dirac:

Yes. Yes. Those were the crucial things that they said.

Wigner:

Not that you are uncommunicative?

Dirac:

No. Why should a factory be concerned whether one is communicative or not?

Kuhn:

What sort of work were you actually doing there?

Dirac:

Well, turning things on lathes and metal work. Filing, drilling.

Kuhn:

Had you had any manual experience of that sort before?

Dirac:

Yes, I had had some in school. This Merchant Venturer School where I went was in the same building as a technical college. It was used as a school in the daytime and a technical college in the evening. It had all the equipment for a technical college, and it was at the disposal of the school. So I actually had metal-working. [Interruption] In Cambridge I came under the influence of R. H. Fowler.... He was my supervisor. Every research student has a supervisor. I came really into closer contact with him than with my Bristol professors, mainly because of the supervision system.... In Cambridge theoretical physics belongs to the mathematics faculty. I was attached to the mathematics faculty. One of the questions which I've always wanted to ask you, Paul, is when did you make a transition from relativity theory to quantum theory?

Dirac:

Well, when I went to Cambridge, I learned about the Bohr theory of the atom from Fowler. I had no idea that atomic theory was so developed. It came as a surprise to me. I had not heard of the Bohr theory of the atom at all in Bristol.

Kuhn:

Had the problem of the quantum arisen there at all? Do you remember?...

Dirac:

I don't think I'd heard about that at all. I was in a mathematics section in Bristol, not in the physics section. They were in different buildings. I didn't have any contact with the physics people.

Kuhn:

Theoretical physics then was not really related with mathematics at Bristol?

Dirac:

At Bristol there was just mathematics, which was divided into the pure and applied parts. I did the applied part. I remember a student, Miss Dent was her name. The first year of the two mathematics years at Bristol we did both pure and applied. The second year I had to choose between them. I wasn't very definite whether I wanted to choose either pure or applied, but Miss Dent definitely wanted to do applied. The professors of course wanted us both to do the same. Otherwise they would have had to give twice as many courses. So by that accident I got shifted into the applied field.

Kuhn:

But the applied mathematics did not include as much theoretical physics as it did at Cambridge?

Dirac:

No, it did not include anything about the Bohr theory of the atom. It was largely about potential theory and solutions of equations, namely $V = 0$.

Wigner:

How, did Fowler get you interested in atomic physics and in quantum theory?

Dirac:

It was his subject.

Wigner:

He talked to you?

Dirac:

Yes, he talked to me about it.

Wigner:

How did he talk? Were you alone when you talked?

Dirac:

I had talks with him alone and also went to his lectures. And, in the lectures I learned about Hamilton methods in dynamics.

Kuhn:

These had also not been part of the applied math?

Dirac:

To some extent I learned about them at Bristol, but I hadn't gone very far with them. It was in Cambridge that I saw the need for the Hamiltonian methods, to connect with the Bohr- Sommerfeld quantization.

Kuhn:

These would have included transformation theories? Hamilton-Jacoby equations?

Dirac:

Yes. Yes. I learned that partly by myself from Whitaker's Analytical Dynamics.... Also Sommerfeld's book. In the appendices, he gives the Hamilton - Jacoby theory.

Wigner:

Haas' book is the one from which I learned it — Theoretical Physics.

Dirac:

I didn't learn it very thoroughly because I remember it was on a Sunday that the idea first occurred to me that ab- ba might correspond. to a Poisson bracket. But at that time I didn't know exactly what a Poisson bracket was, so I wasn't able to check whether it

was right. I didn't have any book at home which dealt with Poisson brackets. I had to wait until the next Monday and go to the library and look up Poisson brackets there and check to see if it was right.

Kuhn:

The ab- ba notion and the relation to Poisson brackets came when in relation to Heisenberg's trip to Cambridge?

Dirac:

Heisenberg made his trip to Cambridge, in I think, June, or it might have been July, of 192. He gave a talk about a new theory to the Kapitza Club, but I wasn't a member of the club so I did not go to the talk. I did not know about it at the time. The first I heard of it was in September when Fowler sent to me a copy of the proofs of Heisenberg's paper and asked me what I thought about it. That was the first that I heard about it. I think Fowler found it interesting. He was a bit uncertain about it and wanted to know what my reaction to it would be. When I first read it I did not appreciate it. I thought there wasn't much in it and I put it aside for a week or so. Then I went back to it later, and suddenly it became clear to me that it was the real thing. And I worked on it intensively starting from September 1925. I think it s just a matter of weeks or so before I got this idea of the Poisson brackets....

Wigner:

I don't think I ever had as extended a conversation with Paul as we are having now, at least not in What was your daily occupation? How much did you go to lectures, how much did you sit in your room, how much did you talk to people? Did you go to theatres?

Dirac:

I never went to theatres. I spent most of my time by myself, sitting working things out or going for walks. I used to spend every Sunday going for a long walk, a whole day walk, taking n lunch with me, like I did yesterday. During those long walks I would not intentionally think about my work, but I might perhaps review it. I found these occasions most profitable for new ideas coming. It was on one of those occasions that the possibility of ab-ba corresponding to a Poisson bracket occurred — on one of those Sunday walks.

Wigner:

But on week days, how much time did you spend in lectures, how, much in your room?

Dirac:

I don't remember just how many lectures I had. maybe four or five a week, something of that order. I might be able to look it up.... I have some notebooks of my lectures. But I would mainly spend the mornings and the evenings studying and took short walks in the afternoons. With a long walk all day Sundays.

Wigner:

Did you have any friends that you saw consistently?

Dirac:

The other research students in my college. I would meet them at dinner every evening.

Wigner:

But not other times too much?

Dirac:

Occasionally I'd be asked to tea, but not very much on that order.

Wigner:

Did you read any literature?

Dirac:

I think I read a little. I don't have anything outstanding in my mind p1' that type.

Wigner:

I have gaps in my knowledge of Paul, and I thought I'd try to fill these.

Kuhn:

I'm very glad you did. Did you also hear Fowler on statistical mechanics?

Dirac:

Yes. I learned about the Boltzmann equation.... Lennard-Jones was there at the time, and he was very much concerned with statistical mechanics.

Wigner:

How did statistical mechanics and quantum mechanics compare in your mind?

Dirac:

I didn't like the statistical mechanics quite so much.

Wigner:

Why not, Paul?

Dirac:

I suppose because of the approximations in it, and the complication of the Boltzmann equation. You have this collision term coming in, which means all the really important things are lumped together in one term, which is not explained very well. I disliked that very much. [Introduction of Mrs. Dirac]

Wigner:

This doesn't agree with the approach of Gibbs.

Dirac:

Yes, I like the Gibbs work very much.... One of r early papers was on statistical mechanics. Not the Gibbs theory. It was a paper which I published in the Royal Society on conditions for statistical equilibrium between atoms, electrons, and radiation. This was before quantum mechanics came out.... My first in the Proceedings.

Kuhn:

That grew out of Fowler's lectures, Lennard-Jones?

Dirac:

Yes, detailed balancing was the subject. It was much talked about then. I wanted to get some information about atomic processes by using this principle of detailed balancing. The adiabatic hypothesis was another of the general principles which one used in those days. I wrote a paper on that.

Kuhn:

Did one generally, in getting into a subject of this sort, get really quite full familiarity with the continental literature also?

Dirac:

Fowler made a visit to Copenhagen, maybe more than one visit. And he'd come back and tell us about what he'd heard in Copenhagen. He inspired in that way. It was through these visits of Fowler to Copenhagen that we were kept in close touch with what the world was doing.... I don't know that I read very much independently of what I got from Fowler. Oh yes, I think I did read Zeitschrift fur Physik quite a lot. That was the main journal in those days. Yes, I read the ZS f. Physik quite a lot. I had learned German at school. Not fluently, but enough to be able to read scientific German at that time. Paul, one thing is more evident to me from your talk than it was before. Even in Bristol, they must have realized very clearly that you have an unusual imagination.

Dirac:

Yes, they did.

Wigner:

Now how did that come if they never saw you alone? Were they (???) 7

Dirac:

Partly from examination results.

Kuhn:

Did you talk to any of them about some of your own thinking on problems of this nature?

Dirac:

No, no. I was very much an introvert. I still am, I expect.... I have heard that the mathematics department at Bristol was very disappointed when they first heard that I had decided to go in for engineering. When I finished my engineering course, they induced me to go study with them for a time. They gave me free tuition.

Wigner:

Whom do you mean by "they" in the mathematics department?

Dirac:

I suppose the professors there. Probably I got that from Fraser; I expect I got that information from 1rar. Fraser was the one that I knew best in the mathematics department at Bristol.

Wigner:

What do you mean by “knew best” if you talked to him so rarely?

Dirac:

I talked to the others still less • But he did invite me to his home once or twice.

Wigner:

How much older was he than you?

Dirac:

Quite a bit older I think, he must have been in his thirties when I was a student....
Everybody agreed that Fraser was a very good mathematics teacher. He didn't interest himself at all in research, but he was a very good teacher.

Kuhn:

You heard lectures of his while you were an engineer?

Dirac:

No, no.

Kuhn:

So your contact with him really came when you transferred to mathematics.

Wigner:

But by that time they must have already known or suspected that you were somewhat out of the ordinary?...

Dirac:

Yes, they must have heard about me previously, probably from examination results.... There was a matriculation examination which I first took when I was slightly less than 1 I took it a second time in more advanced subjects a year later. I'm not sure — I've got the dates wrong. I suppose I ought to check on that. But I did very well in the scientific subjects there.

Wigner:

Just the same there isn't the strength that you exhibit. Examination mainly requires competence.

Dirac:

Weil I don't know how they knew about me. I didn't worry about that.

Wigner:

How did your relation to people change in Cambridge when you first came out with your papers on Poisson brackets?

Dirac:

I don't think my relations to people changed at all. It was just that I became even more concentrated on my work. Oh, there was change in the sense that got elected to societies, the Kapitza Club, the Delta2V Club. I went to their meetings, gave talks to their meetings.

Kuhn:

Did. you find that giving talks on your own work came fairly easily?

Dirac:

Yes. In fact when you've just learned a subject yourself you are in the best position to be able to talk about it because you still remember where the difficulties were. I think you can teach a subject better when you've just learned it than after a number of years. Do you find the same?

Wigner:

Yes, my best class in solid state physics was when I had no idea of the subject, and two of my students were Seitz and Bardeen.... I never found out if they realized that I knew the subject so poorly. I must ask them.

Kuhn:

Besides Fowler in your early days at Cambridge, were there other people with whom you had contact.

Dirac:

There was Cunningham, who taught classical electrodynamics. I went to his lectures. There was Milne who talked about interior of stars. There was Eddington. Eddington was a great man in those days. I introduced relativity to England.

Kuhn:

Was Milne at all concerned with relativity yet then?

Dirac:

I don't think he gave lectures on it, so far as I know.... Just at the moment I can't remember going to Eddington's lectures • Max Newman also gave a course on general relativity, geometrical aspects of the question.

Kuhn:

Were there any of these people whom you also talked with as you did with Fowler?

Dirac:

Not so much. Mime was my supervisor for a term while Fowler was absent in Copenhagen.

Wigner:

I thought Milne was hardly your contemporary.

Dirac:

He was a bit older, a little, probably five years older or so....

Kuhn:

There is an early paper on stellar constitution.

Dirac:

Yes, yes. That was done while I was under Milne's supervision. He suggested the problem.

Kuhn:

I take it from what you say that to a very great extent, what you got from places other than the nature of yourself came from lectures rather than from large amounts of reading.

Dirac:

Well, I think the lectures told me the directions in which I should look, and I fitted in the gaps from reading. As I said before, I would very- seldom hear everything that a lecturer said. Do most people listen to every word that a lecturer says?

Wigner:

I think it is very different if you know the subject reasonably well. Then you can follow every word. If you don't know the subject reasonably well, if it's entirely new, you don't appreciate every word. But didn't you say that you learned mechanics, Hamiltonian theory mainly, from books?

Dirac:

Fowler gave a course of lectures on it also.

Kuhn:

Did you go on with mathematics also at Cambridge, except to the extent that it was integrated into the rest of these subjects?

Dirac:

Only so far as I needed the mathematics for my research.

Kuhn:

Was there considerable excitement about the quantum and the problems of quantum mechanics at Cambridge in this period?

Dirac:

Yes, yes there was.

Kuhn:

Was there also that sense which again people speak of on the continent that something fundamental now had to come to get around these problems that were just not responding. That there was something fundamentally the matter.

Dirac:

I am not sure that that is so. They had the Bohr-Sommerfeld method of quantization and they thought it would have to be extended in some way.... I don't think people suspected that one would need such a complete revolution... It rather came as a surprise to inc when Heisenberg's idea came out....

Wigner:

You know that it was not recognized on the continent that Heisenberg's paper was a fundamental departure. It was thought it might be an improvement of the Schwarzschild-Sommerfeld method of quantization. That it is a. fundamental departure became clear to most of us very much later.

Kuhn:

When you say 'every much later', do you mean more later than the famous three man paper? V

Wigner:

About that time. But even then it was not recognized at all that it would bring the resolution of the fundamental difficulties. For instance the accumulation of energy in the photo-electron, the transfer of angular momentum to the atoms in the Stern-Gerlach experiment, and a number of other similar V basic phenomena. These were considered to be the basic fundamental difficulties. That these were solved was not recognized. It was understood in England, or were these difficulties not so much in the foreground?

Dirac:

I don't know how it was so much with people in general. I just know my own reaction. Just that I had been trying hard for two years to solve a certain problem without any success. Then I suddenly saw that Heisenberg's idea provided the key to the whole mystery.

Kuhn:

What was the problem?

Dirac:

To get a better quantum theory.... Which meant to be able to explain the helium atom.

Wigner:

You did not worry about problems such as how does the energy accumulate in the photo electron?

Dirac:

I tell my primary problem was how to explain the helium spectrum.

Wigner:

Did you read the papers connected with the explanation of the angular momentum transferred to the atom in the Stern-Gerlach experiment?

Dirac:

I remember hearing a lot about the Stern-Gerlach experiment.

Wigner:

There was a paper by Ehrenfest and somebody considering carefully how the angular momentum could be transferred to it so that it becomes quantized.

Dirac:

I don't remember that paper. There was much discussion about the Stern-Gerlach experiment.

Wigner:

This was not the problem which was immediately in your mind, which was foremost in your mind, uppermost in your mind?

Dirac:

It was one of several.

Kuhn:

But your personal worry was really the helium atom?

Dirac:

I don't want to put too much emphasis on it. I was wondering about all the problems together.

Kuhn:

What was it you tried to forge a better quantum mechanics with?

Dirac:

I don't remember in detail. I think I was trying to develop the Bohr-Sommerfeld method of quantization.

Kuhn:

Do you have notebooks? Do you have things that go back to that period?

Dirac:

I used to work on scraps of paper.

Kuhn:

What happened to the scraps of paper?

Dirac:

I've kept a lot of them. I've got some big piles of them.... Most of it is too scrappy to be able to suggest anything. But there may be some bits which would be useful.

Kuhn:

You speak of problems being widely discussed. I take it Fowler clearly was much involved with them himself.

Dirac:

He was the center.... There were not nearly so many students in those days.... Probably only between three and eight.

Kuhn:

When you speak of there being a lot of discussion of the Stern-Gerlach effect, it was in this group of students and Fowler?

Dirac:

I was thinking of meetings of the Delta2V and the Kapitza Club.

Kuhn:

How big were those meetings? About 20.... They were mainly students, and faculty would come when they wanted to. The Kapitza Club met every week, and the Delta2 v, I think it was twice a term. It was mainly from these meetings of clubs rather than just discussion groups. The colleges also had their mathematical societies. The undergraduates would get someone to talk. Maybe one of them, or maybe one of the faculty come talk to them.

Kuhn:

Did quantum mechanics become large in these discussions as compared with relativity?

Dirac:

Yes. The Kapitza Club has kept a minute book. I don't know where it is now. You should get hold of that minute book of the Kapitza Club, that would be a list of all the early talks. It's stopped meeting for the last two or three years. But you ought to get hold of that minute book.